A Full Proofs

Theorem 3.7. *For any sequent Ω, if it is derivable in $\mathsf{SC}_{\mathbf{FIK}}$, then it is valid in* **FIK**.

Proof. We prove this by induction on the structure of a derivation. It suffices to verify that, the initial sequents are valid and the inference rules preserve validity. For the rules, we only show the non-trivial cases, namely $(\supset_2^\bullet), (\supset_2^\circ), (\Box_2^\circ)$ and $(\Diamond_2^\bullet)$.

$(\supset_2^\bullet)$ We only show the case when $F^\circ \in \Phi$ as the other cases are similar. Assume w.l.o.g. that Ω is block-free and the output formula of the conclusion is in Ω. Then the rule is of the following form

$$\frac{\Omega^*, [\Phi^*, A \supset B^\bullet, A^\circ] \qquad \Omega^*, [\Phi, B^\bullet], F^\circ}{\Omega^*, [\Phi, A \supset B^\bullet], F^\circ}\ (\supset_2^\bullet)$$

Assume this rule does not preserve validity, then there exists a bi-relational model $\mathscr{M} = (W, \leq, R, V)$ with $x \in W$ such that $x \nVdash \Omega^*, [\Phi, A \supset B^\bullet], F^\circ$. By definition, there is $x' \in W$ such that $x \leq x'$ and $x' \Vdash \bigwedge \Omega \wedge \Diamond(\bigwedge \Phi \wedge (A \supset B))$ and $x' \nVdash F$. It follows that there is $y \in W$ such that $y \Vdash (A \supset B) \wedge \bigwedge \Phi$. Since the left premise is valid, we have $x' \Vdash \Omega^*, [\Phi^*, A \supset B^\bullet, A^\circ]$. Thus $y \Vdash A$. Recall $y \Vdash A \supset B$, so $y \Vdash B$. Meanwhile, the right premise is also valid, so $x' \Vdash \Omega^*, [\Phi, B^\bullet], F^\circ$, which implies $x' \nVdash \Diamond(\bigwedge \Phi \wedge B)$. Since $Rx'y$ and $y \Vdash \bigwedge \Phi$, we have $y \nVdash B$, a contradiction.

$(\supset_2^\circ)$ Assume w.l.o.g. that Ω is block-free and this rule does not preserve validity, then there exists a bi-relational model $\mathscr{M} = (W, \leq, R, V)$ with $x \in W$ such that $x \nVdash \Omega, [\Phi, A \supset B^\circ]$. By definition, there are x', y_0 such that $x \leq x', Rx'y, x' \Vdash \bigwedge \Omega, y_0 \Vdash \bigwedge \Phi$ and $y_0 \nVdash A \supset B$. This implies there are $y' \geq y_0$ such that $y' \Vdash A$ and $y' \nVdash B$. Moreover, $y' \Vdash \bigwedge \Phi$. Thus $y' \nVdash \Phi, A^\bullet, B^\circ$. Since the premise is valid, $y' \Vdash \Phi, A^\bullet, B^\circ$, a contradiction.

$(\Box_2^\circ)$ Assume w.l.o.g. that Ω is block-free and this rule does not preserve validity, then there exists a bi-relational model $\mathscr{M} = (W, \leq, R, V)$ with $x \in W$ such that $x \nVdash \Omega, [\Phi, \Box A^\circ]$. By definition, there are $x', y \in W$ such that $x' \geq x, Rx'y, x' \Vdash \bigwedge \Omega, y \Vdash \bigwedge \Phi$ and $y \nVdash \Box A$. This implies that there is $u, v \in W$ such that $u \geq y$ and Ruv and $v \nVdash A$. Since the premise is valid, we have $u \Vdash \Phi, [A^\circ]$, so $u \nVdash \bigwedge \Phi$. Meanwhile, since $y \Vdash \bigwedge \Phi$ and $u \geq y$, by monotonicity, $u \Vdash \bigwedge \Phi$, a contradiction.

$(\Diamond_2^\bullet)$ The rule is of one of the two forms:

$$\frac{\Phi, [A^\bullet], F^\circ}{\Omega, [\Phi, \Diamond A^\bullet, F^\circ]} \qquad \frac{\Phi, [A^\bullet], \bot^\circ}{\Omega, [\Phi, \Diamond A^\bullet]}$$

For the first case, assume this rule does not preserve validity, then there exists a bi-relational model $\mathscr{M} = (W, \leq, R, V)$ with $x \in W$ such that $x \nVdash \Omega, [\Phi, \Diamond A^\bullet, F^\circ]$. By definition, this means there is $x', y \in W$ such that $x' \geq x, Rx'y$ and $y \Vdash \bigwedge \Phi \wedge \Diamond A$ and $y \nVdash F$. Meanwhile, since the premise is valid, we have $y \Vdash \Phi, [A^\bullet], F^\circ$, which means $y \Vdash \bigwedge \Phi \wedge \Diamond A \supset F$. Since $y \leq y$, by $y \Vdash \bigwedge \Phi \wedge \Diamond A$ it follows $y \Vdash F$, a contradiction.

For the second case, the out put formula occurs in Ω. Assume this rule does not preserve validity, then there exists a bi-relational model $\mathscr{M} = (W, \leq, R, V)$ with $x \in W$ such that $x \nVdash \Omega, [\Phi, \Diamond A^\bullet]$. By definition, regardless of the exact location of the output formula, there exists $x' \in W$ such that $x' \geq x$ such that $x' \Vdash \Diamond(\bigwedge \Phi \wedge \Diamond A)$. This further implies there is $y \in W$ such that $Rx'y$ and $y \Vdash \bigwedge \Phi \wedge \Diamond A$. Meanwhile, since the premise is valid, we have $y \Vdash \Phi, [A^\bullet], \bot^\circ$, which means $y \nVdash \bigwedge \Phi \wedge \Diamond A$, a contradiction.

This completes the proof. $\square$

Proposition 4.6. *The following structural rules are height-preserving admissible in* $\mathsf{SC}_{\mathbf{FIK}}$,

$$\frac{A^\bullet, A^\bullet, \Omega}{A^\bullet, \Omega}\,(c_1^\bullet) \qquad \frac{\Omega, [A^\bullet, A^\bullet, \Phi]}{\Omega, [A^\bullet, \Phi]}\,(c_2^\bullet) \qquad \frac{\Omega, [\Phi], [\Phi^*]}{\Omega, [\Phi]}\,(c_{[\cdot]}^-) \qquad \frac{\Phi}{\Omega, [\Phi]}\,([\cdot])$$

Proof. For the first six rules, the proof is standard by induction on the height of a derivation. We only provide the proof for $(c_{[\cdot]}^-)$ here. Assume the upper sequent $\Omega, [\Phi], [\Phi^*]$ is derived in a derivation with the last rule (r). The only non-trivial case is when $r = \supset_2^\bullet$ and is applied to Φ^* while the output formula is in Φ. Assume $\Phi = \Phi_0, A \supset B^\bullet, F^\circ$, then $\Phi^* = \Phi_0^*, A \supset B^\bullet$. In this case we have the following transformation of the derivation

$$\dfrac{\dfrac{\Omega, [\Phi_0, A \supset B^\bullet], [\Phi_0, A \supset B^\bullet, A^\circ] \qquad \Omega, [\Phi_0, A \supset B^\bullet, F^\circ], [\Phi_0, B^\bullet]}{\Omega, [\Phi_0, A \supset B^\bullet, F^\circ], [\Phi_0, A \supset B^\bullet]}\,(\supset_2^\bullet)}{\Omega, [\Phi_0, A \supset B^\bullet, F^\circ]} \quad \rightsquigarrow \quad \dfrac{\dfrac{\Omega, [\Phi_0, A \supset B^\bullet], [\Phi_0, A \supset B^\bullet, A^\circ]}{\Omega, [\Phi_0, A \supset B^\bullet, A^\circ]}\,\text{IH} \qquad \dfrac{\dfrac{\Omega, [\Phi_0, A \supset B^\bullet, F^\circ], [\Phi_0, B^\bullet]}{\Omega, [\Phi_0, B^\bullet, F^\circ], [\Phi_0, B^\bullet]}\,\text{inver-}\supset_2^\bullet}{\Omega, [\Phi_0, B^\bullet, F^\circ]}\,\text{IH}}{\Omega, [\Phi_0, A \supset B^\bullet, F^\circ]}$$

For $([\cdot])$, notice that according to the form of the rule, Ω on the conclusion can only be an input sequent. Assume Φ is derived from some Ψ by a rule (r), then (r) can only be a 1-rule. To obtain the conclusion, we take the 2-rule counterpart (r') of (r) and apply (r') to Ψ with the context Ω. For example, assume $r = \Diamond_1^\bullet$ and $\Phi = \Phi', \Diamond A^\bullet$. In this case we have the following transformation of the derivation:

$$\dfrac{\dfrac{\Phi', [A^\bullet]}{\Phi', \Diamond A^\bullet}\,\Diamond_1^\bullet}{\Omega, [\Phi', \Diamond A^\bullet]} \quad \rightsquigarrow \quad \dfrac{\Phi', [A^\bullet]}{\Omega, [\Phi', \Diamond A^\bullet]}\,\Diamond_2^\bullet$$

This completes the proof. □

Theorem 4.9 (Cut-admissibility)**.** *Both* (cut) *and* $(\mathsf{cut}^{[\cdot]})$ *are admissible in* $\mathsf{SC}_{\mathbf{FIK}}$*, meaning that if a sequent S is derivable in* $\mathsf{SC}_{\mathbf{FIK}} + \mathsf{cut} + \mathsf{cut}^{[\cdot]}$ *then it is also derivable in* $\mathsf{SC}_{\mathbf{FIK}}$.

Proof. Let us consider all the cases of (D1)-(D4) and (E1)-(E4) in turn.
(D1) one of the premises is an axiom, which implies one of the following: (1) Ω^* or Ω is an axiom, so is the sequent on the conclusion; (2) $A = p$ and $p^\bullet \in \Omega$, then we obtain Ω just by contraction from the right premise $A^\bullet, \Omega$; (3) $A = p$ and $p^\circ \in \Omega$, dual to the second case.
(D2) cut formula A is principal in both premises, we consider the following non-trivial sub-cases:

- If $A = B \supset C$, we have

$$\dfrac{\dfrac{\begin{array}{c}\mathscr{D}_1'\\ \Omega^*, B^\bullet, C^\circ\end{array}}{\Omega^*, B \supset C^\circ}\,(\supset^\circ) \qquad \dfrac{\begin{array}{c}\mathscr{D}_2'\\ B \supset C^\bullet, \Omega^*, B^\circ\end{array} \qquad \begin{array}{c}\mathscr{D}_2''\\ C^\bullet, \Omega\end{array}}{B \supset C^\bullet, \Omega}\,(\supset^\bullet)}{\Omega}\,(\mathsf{cut}) \quad \rightsquigarrow \quad \dfrac{\dfrac{\dfrac{\Omega^*, B \supset C^\circ \qquad B \supset C^\bullet, \Omega^*, B^\circ}{\Omega^*, B^\circ}\,(\mathsf{cut}) \qquad \Omega^*, B^\bullet, C^\circ}{\Omega^*, C^\circ}\,(\mathsf{cut}) \qquad C^\bullet, \Omega}{\Omega}\,(\mathsf{cut})$$

- If $A = \Box B$, further assume $\Omega = \Omega', [\Phi]$, then we have

$$\dfrac{\dfrac{\begin{array}{c}\mathscr{D}_1'\\ \Omega^*, [B^\circ]\end{array}}{\Omega^*, \Box B^\circ}\,(\Box^\circ) \qquad \dfrac{\begin{array}{c}\mathscr{D}_2'\\ \Omega', \Box B^\bullet, [B^\bullet, \Phi]\end{array}}{\Omega', \Box B^\bullet, [\Phi]}\,(\Box^\bullet)}{\Omega', [\Phi]}\,(\mathsf{cut}) \quad \rightsquigarrow \quad \dfrac{\Omega^*, [B^\circ] \qquad \dfrac{\dfrac{\Omega'^*, [\Phi^*], \Box B^\circ}{\Omega'^*, [B^\bullet, \Phi^*], \Box B^\circ}\,(w_{[\cdot]}^\bullet) \qquad \Omega', \Box B^\bullet, [B^\bullet, \Phi]}{\Omega', [B^\bullet, \Phi]}\,(\mathsf{cut})}{\Omega, [\Phi]}\,(\mathsf{cut}^{[\cdot]})$$

- If $A = \Diamond B$, assume $\Omega = \Omega', [\Phi]$ and the output formula of the conclusion is in Φ, then we have

$$\dfrac{\dfrac{\begin{array}{c}\mathscr{D}'_1\\ \Omega'^*,[\Phi^*,B^\circ]\end{array}}{\Omega'^*,[\Phi^*],\Diamond B^\circ}(\Diamond^\circ) \quad \dfrac{\begin{array}{c}\mathscr{D}'_2\\ \Omega',[\Phi],[B^\bullet]\end{array}}{\Diamond B^\bullet,\Omega',[\Phi]}(\Diamond^\bullet)}{\Omega',[\Phi]}(\mathsf{cut}) \quad \rightsquigarrow \quad \dfrac{\dfrac{\dfrac{\Omega'^*,[\Phi^*,B^\circ]}{\Omega'^*,[\Phi^*,B^\circ],[\Phi^*]}(w^+_{[\cdot]}) \quad \dfrac{\Omega',[\Phi],[B^\bullet]}{\Omega',[\Phi],[\Phi^*,B^\bullet]}(w^\bullet)}{\Omega',[\Phi^*],[\Phi]}(\mathsf{cut}^{[\cdot]})}{\Omega',[\Phi]}w^-_{[\cdot]}$$

The other case when the output formula is in Ω' is similar.

(D3) A° is principal in (r_1) while $A^\bullet$ is side in (r_2). We have the following sub-cases:

(i) r_2 is one of the invertible rules except $\Box^\circ_1$. As an example, we show the case when $r_2 = \supset^\bullet_1$. Suppose $\Omega = B \supset C^\bullet, \Omega'$, in this case, we have the following derivation unfolding $\mathscr{D}$,

$$\dfrac{\begin{array}{c}\mathscr{D}'_1\\ B\supset C^\bullet,\Omega'^*,A^\circ\end{array} \quad \dfrac{\begin{array}{c}\mathscr{D}'_{21}\\ A^\bullet,B\supset C^\bullet,\Omega'^*,B^\circ\end{array} \quad \begin{array}{c}\mathscr{D}'_{22}\\ A^\bullet,C^\bullet,\Omega'\end{array}}{A^\bullet,B\supset C^\bullet,\Omega'}(\supset^\bullet_1)}{B\supset C^\bullet,\Omega'^*}(\mathsf{cut})$$

Then we transform the derivation into the following

$$\dfrac{\dfrac{\dfrac{\dfrac{B\supset C^\bullet,\Omega'^*,A^\circ}{B\supset C^\bullet,B\supset C^\bullet,\Omega'^*,A^\circ}(w^\bullet) \quad \dfrac{A^\bullet,B\supset C^\bullet,\Omega'^*,B^\circ}{A^\bullet,B\supset C^\bullet,B\supset C^\bullet,\Omega'^*,B^\circ}(w^\bullet)}{B\supset C^\bullet,\Omega'^*,B^\circ}(\mathsf{cut}) \quad \dfrac{\dfrac{B\supset C^\bullet,\Omega'^*,A^\circ}{C^\bullet,B\supset C^\bullet,\Omega'^*,A^\circ}(w^\bullet) \quad \dfrac{A^\bullet,C^\bullet,\Omega'}{A^\bullet,C^\bullet,B\supset C^\bullet,\Omega'}(w^\bullet)}{C^\bullet,B\supset C^\bullet,\Omega'}(\mathsf{cut})}{B\supset C^\bullet,B\supset C^\bullet,\Omega'}(\supset^\bullet_1)}{B\supset C^\bullet,\Omega'}(w^\bullet)$$

(ii) suppose $r_2 = \Box^\circ_1$ and $\Omega = \Omega', \Box B^\circ$. Hence $\Omega^* = \Omega'$ and the right premise of cut is derived from $A^\bullet, \Omega', [B^\circ]$. We have the following sub-cases according to the shape of A,

- $A = C \vee D$ and $r_1 = \vee^\circ$, in this case, the left premise of (cut) is $\Omega_1, C\vee D^\circ$ and is derived from Ω_1, C°, the right premise of (cut) is $C\vee D^\bullet, \Omega'_2, \Box B^\circ$. By Lemma 4.5, both $C^\bullet, \Omega'_2, \Box B^\circ$ and $D^\bullet, \Omega'_2, \Box B^\circ$ are derivable at the same height. Then the proof proceeds in the same way as in (D2). $A = C \wedge D$ is similar.
- $A = C \supset D$ and $r_1 = \supset^\circ_1$, we apply (cut) to the premise of $(\Box^\circ_1)$

$$\dfrac{\dfrac{\dfrac{\begin{array}{c}\mathscr{D}'_1\\ \Omega',C^\bullet,D^\circ\end{array}}{\Omega',C\supset D^\circ}(\supset^\circ_1) \quad \begin{array}{c}\mathscr{D}'_2\\ C\supset D^\bullet,\Omega',[B^\circ]\end{array}}{\Omega',[B^\circ]}(\mathsf{cut})}{\Omega',\Box B^\circ}(\Box^\circ_1)$$

- $A = \Box C$ and $r_1 = \Box^\circ_1$, we have

$$\dfrac{(r_1)\dfrac{\begin{array}{c}\mathscr{D}'_1\\ \Omega',[C^\circ]\end{array}}{\Omega',\Box C^\circ} \quad \dfrac{\begin{array}{c}\mathscr{D}'_2\\ \Box C^\bullet,\Omega',[B^\circ]\end{array}}{\Box C^\bullet,\Omega',\Box B^\circ}(\Box^\circ_1)}{\Omega',\Box B^\circ}(\mathsf{cut}) \quad \rightsquigarrow \quad \dfrac{\dfrac{\Omega',[C^\circ] \quad \dfrac{\Omega',\Box C^\circ \quad \Omega',\Box C^\bullet,[C^\bullet,B^\circ]}{\Omega',[C^\bullet,B^\circ]}(\mathsf{cut})}{\Omega',[B^\circ]}(\mathsf{cut}^{[\cdot]})}{\Omega',\Box B^\circ}(\Box^\circ_1)$$

- $A = \Diamond C, r_1 = \Diamond_1^\circ$ and $\Omega' = \Omega'', [\Phi]$, since $\Omega = \Omega', \Box B^\circ$, we have $\Omega^* = \Omega'', [\Phi]$.

$$\dfrac{(r_1)\dfrac{\overset{\mathscr{D}_1'}{\vdots}\ \Omega'', [\Phi, C^\circ]}{\Omega'', \Diamond C^\circ, [\Phi]} \quad \dfrac{\overset{\mathscr{D}_2'}{\vdots}\ \Diamond C^\bullet, \Omega'', [\Phi], [B^\circ]}{\Diamond C^\bullet, \Omega'', [\Phi], \Box B^\circ}(\Box_1^\circ)}{\Omega'', [\Phi], \Box B^\circ}(\mathsf{cut}) \quad \rightsquigarrow \quad \dfrac{\dfrac{\dfrac{\Omega'', [\Phi, C^\circ]}{\Omega'', [\Phi], [\Phi, C^\circ]}(w^+_{[\cdot]}) \quad \dfrac{\Omega'', [\Phi], \Box B^\circ, [C^\bullet]}{\Omega'', [\Phi], \Box B^\circ, [C^\bullet, \Phi]}(w^\bullet)}{\Omega'', [\Phi], \Box B^\circ, [\Phi]}(\mathsf{cut}^{[\cdot]})}{\Omega'', [\Phi], \Box B^\circ}(c^-_{[\cdot]})$$

(iii) r_2 is one of the non-invertible rules, according to the form of rules, we see $r_2 \in \{\Diamond^\circ, \vee_1^\circ, \vee_2^\circ\}$. Among all these three cases can be done by switching the order of (e_2) and (cut), we only show the case when $r_2 = \vee_2^\circ$ as an example. Assume $\Omega = \Omega', [\Phi, B \vee C^\circ]$. We have

$$\dfrac{(e_1)\dfrac{\overset{\mathscr{E}_1'}{\vdots}\ \Omega', [\Phi], A^\circ}{} \quad \dfrac{\overset{\mathscr{E}_2'}{\vdots}\ A^\bullet, \Omega', [\Phi, B^\circ]}{A^\bullet, \Omega', [\Phi, B \vee C^\circ]}(\vee_2^\circ)}{\Omega', [\Phi, B \vee C^\circ]}(\mathsf{cut}^{[\cdot]}) \quad \rightsquigarrow \quad \dfrac{\dfrac{\Omega', [\Phi], A^\circ \qquad A^\bullet, \Omega', [\Phi, B^\circ]}{\Omega', [\Phi, B^\circ]}(\mathsf{cut})}{\Omega', [\Phi, B \vee C^\circ]}(\vee_2^\circ)$$

(D4) cut formula A° is *side* in the left premise. This further implies the last rule (r_1) applied in $\mathscr{D}_1$ is one of the invertible $\bullet$-rules since other rules do not have side $\circ$-formulas (outside a block) on the conclusion. The proof is the same as the *side* case (i) covered in **(D3)**.
(E1) one of the premises is an axiom, similar to the corresponding cases in (D1);
(E2) cut formula A is principal in both premises. Note that Φ is a simple sequent, namely block-free, so A cannot be of the form $\Box B$ or $\Diamond B$ under the assumption of (E2). $A = B \vee C$ or $B \wedge C$ is trivial, we only consider $A = B \supset C$, we have

$$\dfrac{\dfrac{\overset{\mathscr{E}_1'}{\vdots}\ \Phi^*, B^\bullet, C^\circ}{\Omega^*, [\Phi^*, B \supset C^\circ]}(\supset_2^\circ) \quad \dfrac{\overset{\mathscr{E}_2'}{\vdots}\ \Omega^*, [B \supset C^\bullet, \Phi^*, B^\circ] \quad \overset{\mathscr{E}_2''}{\vdots}\ \Omega, [C^\bullet, \Phi]}{\Omega, [B \supset C^\bullet, \Phi]}(\supset_2^\bullet)}{\Omega, [\Phi]}(\mathsf{cut}^{[\cdot]}) \quad \rightsquigarrow \quad \dfrac{\dfrac{\dfrac{\Omega^*, [\Phi^*, B \supset C^\circ] \quad \Omega^*, [B \supset C^\bullet, \Phi^*, B^\circ]}{\Omega^*, [\Phi^*, B^\circ]}(\mathsf{cut}^{[\cdot]}) \quad \dfrac{\Phi^*, B^\bullet, C^\circ}{\Omega^*, [\Phi^*, B^\bullet, C^\circ]}([\cdot])}{\Omega^*, [\Phi^*, C^\circ]}(\mathsf{cut}^{[\cdot]}) \quad \Omega, [C^\bullet, \Phi]}{\Omega, [\Phi]}(\mathsf{cut}^{[\cdot]})$$

(E3) A° is principal in (e_1) while $A^\bullet$ is *side* in (e_2). In this case, let us discuss different shapes of A. If $A = B \vee C$ or $B \wedge C$, then it follows directly by invertibility of $A^\bullet$ in the right premise of cut. Moreover, a $\Diamond^\circ$-formula cannot be principal in a block, hence we only need to consider the case when $A = C \supset D$ and $A = \Box C$. We further discuss the possibilities of r_2. Then the rule (e_2) is among $(\vee_2^\circ)(\vee_1^\circ)(\Box_2^\circ)(\supset_2^\circ)(\Diamond_2^\bullet)$ and all the invertible rules. Thus we have the following sub-cases,

(i) e_2 is one of the invertible rules, then the proof proceeds in the same way as in **(D3)**.

(ii) e_2 is one of the non-invertible rules and $e_2 \in \{\vee_2^\circ, \vee_1^\circ, \Box_2^\circ, \supset_2^\circ, \Diamond_2^\bullet\}$ applied to some $F^\circ \in \Phi$. When e_2 is one of these $\circ$-rules, we write $\Phi = \Phi', F^\circ$, hence $\Omega^* = \Omega$ and $\Phi^* = \Phi'$. We only present the cases when $A = C \supset D$ as the other one $A = \Box C$ is similar.

(1) $e_2 = \Box_2^\circ$ and $\Phi = \Phi', \Box B^\circ$. In this case, we have

$$\dfrac{(e_1)\dfrac{\overset{\mathscr{E}_1'}{\vdots}\ \Phi', C^\bullet, D^\circ}{\Omega, [\Phi', C \supset D^\circ]} \quad \dfrac{\overset{\mathscr{E}_2'}{\vdots}\ C \supset D^\bullet, \Phi', [B^\circ]}{\Omega, [C \supset D^\bullet, \Phi', \Box B^\circ]}(\Box_2^\circ)}{\Omega, [\Phi', \Box B^\circ]}(\mathsf{cut}) \quad \rightsquigarrow \quad \dfrac{\dfrac{\dfrac{\overset{\mathscr{E}_1'}{\vdots}\ \Phi', C^\bullet, D^\circ}{\Phi', C \supset D^\circ}(\supset_1^\circ) \quad \overset{\mathscr{E}_2'}{\vdots}\ C \supset D^\bullet, \Phi', [B^\circ]}{\Phi', [B^\circ]}(\mathsf{cut})}{\Omega, [\Phi', \Box B^\circ]}(\Box_2^\circ)$$

(2) $e_2 = \supset_2^\circ$ and $\Phi = \Phi', E \supset F^\circ$. Similar as above.

(3) $e_2 = \vee_2^\circ$ or $\vee_1^\circ$, we only show the case when $e_2 = \vee_2^\circ$, the other is similar. Assume $\Phi = \Phi', B_1 \vee B_2^\circ$, we have $A = C \supset D$ and $e_1 = \supset_2^\circ$, we have

$$\dfrac{(e_1)\dfrac{\begin{array}{c}\mathscr{E}_1'\\ \Phi', C^\bullet, D^\circ\end{array}}{\Omega, [\Phi', C \supset D^\circ]} \quad \dfrac{\begin{array}{c}\mathscr{E}_2'\\ \Omega, [C \supset D^\bullet, \Phi', B_1^\circ]\end{array}}{\Omega, [C \supset D^\bullet, \Phi', B_1 \vee B_2^\circ]}(\vee_2^\circ)}{\Omega, [\Phi', B_1 \vee B_2^\circ]}(\mathsf{cut}) \quad \rightsquigarrow \quad \dfrac{\dfrac{\dfrac{\begin{array}{c}\mathscr{E}_1'\\ \Phi', C^\bullet, D^\circ\end{array}}{\Omega, [\Phi', C \supset D^\circ]}(\supset_2^\circ) \quad \begin{array}{c}\mathscr{E}_2'\\ \Omega, [C \supset D^\bullet, \Phi', B_1^\circ]\end{array}}{\Omega, [\Phi', B_1^\circ]}(\mathsf{cut})}{\Omega, [\Phi', B_1 \vee B_2^\circ]}(\vee_2^\circ)$$

(4) $e_2 = \Diamond_2^\bullet$ and $\Phi = \Diamond B^\bullet, \Phi'$. In this case, we have

$$\dfrac{(\supset_2^\circ)\dfrac{\begin{array}{c}\mathscr{E}_1'\\ \Diamond B^\bullet, \Phi'^*, C^\bullet, D^\circ\end{array}}{\Omega^*, [\Diamond B^\bullet, \Phi'^*, C \supset D^\circ]} \quad \dfrac{\begin{array}{c}\mathscr{E}_2'\\ C \supset D^\bullet, \Phi'^+, [B^\bullet]\end{array}}{\Omega, [C \supset D^\bullet, \Diamond B^\bullet, \Phi']}(\Diamond_2^\bullet)}{\Omega, [\Diamond B^\bullet, \Phi']}(\mathsf{cut}^{[\cdot]}) \quad \rightsquigarrow \quad \dfrac{\dfrac{\dfrac{\dfrac{\begin{array}{c}\mathscr{E}_1'\\ \Diamond B^\bullet, \Phi'^*, C^\bullet, D^\circ\end{array}}{\Diamond B^\bullet, \Phi'^*, C \supset D^\circ}(\supset_1^\circ)}{\Diamond B^\bullet, \Phi'^*, [B^\bullet], C \supset D^\circ}(w_{[\cdot]}^+) \quad \dfrac{\begin{array}{c}\mathscr{E}_2'\\ C \supset D^\bullet, \Phi'^+, [B^\bullet]\end{array}}{C \supset D^\bullet, \Diamond B^\bullet, \Phi'^+, [B^\bullet]}(w^\bullet)}{\Diamond B^\bullet, \Phi', [B^\bullet]}(\mathsf{cut}^{[\cdot]})}{\Omega, [\Diamond B^\bullet, \Diamond B^\bullet, \Phi']}(\Diamond_2^\bullet)}{\Omega, [\Diamond B^\bullet, \Phi']}(c^\bullet)$$

(E4) A° is *side* in (e_1). This implies the last rule (e_1) applied in $\mathscr{E}_1$ is one of the $\bullet$-invertible rules or is $(\Diamond_2^\bullet)$. For the invertible rules, the proof is similar to the *side* case (i) covered in **(D3)**. Here we only show the case when $e_1 = \Diamond_2^\bullet$. In this case, $\Phi = \Diamond B^\bullet, \Phi'$ and derivation $\mathscr{E}$ is unfolded as

$$\dfrac{(e_1)\dfrac{\begin{array}{c}\mathscr{E}_1'\\ \vdots\end{array}}{\Omega^*, [\Phi'^*, \Diamond B^\bullet, A^\circ]}(e_1') \quad \dfrac{\begin{array}{c}\mathscr{E}_2'\\ \vdots\end{array}}{\Omega, [A^\bullet, \Phi', \Diamond B^\bullet]}(e_2)}{\Omega, [\Phi', \Diamond B^\bullet]}(\mathsf{cut}^{[\cdot]})$$

We discuss the role of $A^\bullet$ in (e_2) and the role of A° in (e_1'). As explained in **(D4)**, A° cannot be weak in (e_1'), hence we have the following cases:

(i) $A^\bullet$ is *weak* in (e_2), trivial.

(ii) both $A^\bullet$ and A° are *principal*, then according to the form of rules, A cannot be a $\Box$-formula. Thus we have the following sub-cases:

- $A = C \wedge D$ or $C \vee D$. We only show the first one. In this case, we have

$$\dfrac{(e_1)\dfrac{(e_1')\dfrac{\begin{array}{c}\mathscr{E}_{11}'\\ \Phi'^*, [B^\bullet], C^\circ\end{array} \quad \begin{array}{c}\mathscr{E}_{12}'\\ \Phi'^*, [B^\bullet], D^\circ\end{array}}{\Phi'^*, [B^\bullet], C \wedge D^\circ}}{\Omega^*, [\Phi'^*, \Diamond B^\bullet, C \wedge D^\circ]} \quad \dfrac{\begin{array}{c}\mathscr{E}_2'\\ \Omega, [C^\bullet, D^\bullet, \Phi', \Diamond B^\bullet]\end{array}}{\Omega, [C \wedge D^\bullet, \Phi', \Diamond B^\bullet]}(e_2)}{\Omega, [\Phi', \Diamond B^\bullet]}(\mathsf{cut}^{[\cdot]}) \quad \rightsquigarrow \quad \dfrac{\dfrac{\Phi'^*, [B^\bullet], D^\circ}{\Omega^*, [\Phi'^*, \Diamond B^\bullet, D^\circ]}(\Diamond_2^\bullet) \quad \dfrac{\dfrac{\dfrac{\Phi'^*, [B^\bullet], C^\circ}{\Omega^*, [\Phi'^*, \Diamond B^\bullet, C^\circ]}(\Diamond_2^\bullet)}{\Omega^*, [\Phi'^*, \Diamond B^\bullet, D^\bullet, C^\circ]}(w^\bullet) \quad \Omega, [C^\bullet, D^\bullet, \Phi', \Diamond B^\bullet]}{\Omega, [D^\bullet, \Phi', \Diamond B^\bullet]}(\mathsf{cut}^{[\cdot]})}{\Omega, [\Phi', \Diamond B^\bullet]}(\mathsf{cut}^{[\cdot]})$$

- $A = C \supset D$. In this case, derivation $\mathscr{E}$ is unfolded as

$$\dfrac{(e_1)\dfrac{(e_1')\dfrac{\begin{array}{c}\mathscr{E}_1'\\ \Phi'^*, [B^\bullet], C^\bullet, D^\circ\end{array}}{\Phi'^*, [B^\bullet], C \supset D^\circ}}{\Omega^*, [\Phi'^*, \Diamond B^\bullet, C \supset D^\circ]} \quad \dfrac{\begin{array}{c}\mathscr{E}_{21}'\\ \Omega^*, [C \supset D^\bullet, \Phi'^*, \Diamond B^\bullet, C^\circ]\end{array} \quad \begin{array}{c}\mathscr{E}_{22}'\\ \Omega, [D^\bullet, \Phi', \Diamond B^\bullet]\end{array}}{\Omega, [C \supset D^\bullet, \Phi', \Diamond B^\bullet]}(e_2)}{\Omega, [\Phi', \Diamond B^\bullet]}(\mathsf{cut}^{[\cdot]})$$

and we transform it into

$$\dfrac{\dfrac{\dfrac{\Omega^*,[\Phi'^*,\Diamond B^\bullet,C\supset D^\circ] \qquad \Omega^*,[C\supset D^\bullet,\Phi'^*,\Diamond B^\bullet,C^\circ]}{\Omega^*,[\Phi'^*,\Diamond B^\bullet,C^\circ]}\,(\mathsf{cut}^{[\cdot]}) \qquad \dfrac{\Phi'^*,[B^\bullet],C^\bullet,D^\circ}{\Omega^*,[\Phi'^*,\Diamond B^\bullet,C^\bullet,D^\circ]}\,(\Diamond_2^\bullet)}{\Omega^*,[\Phi'^*,\Diamond B^\bullet,D^\circ]}\,(\mathsf{cut}^{[\cdot]}) \qquad \Omega,[D^\bullet,\Phi',\Diamond B^\bullet]}{\Omega,[\Phi',\Diamond B^\bullet]}\,(\mathsf{cut}^{[\cdot]})$$

- $A=\Diamond C$. In this case, we have

$$\dfrac{\dfrac{\dfrac{\overset{\mathscr{E}_1'}{\nabla}}{\Phi'^*,[B^\bullet,C^\circ]}\,(e_1')}{\Omega^*,[\Phi'^*,\Diamond B^\bullet,\Diamond C^\circ]}\,(e_1) \qquad \dfrac{\overset{\mathscr{E}_2'}{\nabla}}{\Omega,[\Diamond C^\bullet,\Phi',\Diamond B^\bullet]}\,(e_2)}{\Omega,[\Phi',\Diamond B^\bullet]}\,(\mathsf{cut}^{[\cdot]}) \quad\rightsquigarrow\quad \dfrac{\dfrac{\dfrac{\dfrac{\Phi'^*,[B^\bullet,C^\circ]}{\Phi'^*,[B^\bullet,C^\circ],\Diamond B^\bullet}\,(w^\bullet) \qquad \dfrac{[C^\bullet],\Phi'^+,\Diamond B^\bullet}{[C^\bullet,B^\bullet],\Phi'^+,\Diamond B^\bullet}\,(w^\bullet)}{[B^\bullet],\Phi,\Diamond B^\bullet}\,(\mathsf{cut}^{[\cdot]})}{\Omega,[\Diamond B^\bullet,\Phi',\Diamond B^\bullet]}\,(\Diamond_2^\bullet)}{\Omega,[\Phi',\Diamond B^\bullet]}\,(c^\bullet)$$

(iii) $A^\bullet$ is *side* in (e_2), then according to the form of rules, we see e_2 can be any rule of the calculus.

- if e_2 is an invertible rule or $\Diamond^\circ,\vee^\circ$, we switch the order of (e_2) and $(\mathsf{cut}^{[\cdot]})$, that is, apply $(\mathsf{cut}^{[\cdot]})$ to the premise of (e_2) and the conclusion of (e_1') first, then apply (e_2).
- if $e_2\in\{\supset_2^\circ,\Box_2^\circ,\Diamond_2^\bullet\}$, the three cases are similar and we only show one here. Assume $\Phi'=\Psi,C\supset D^\circ$, then $\Omega^*=\Omega$, $\Phi'^*=\Psi$, and we have

$$\dfrac{\dfrac{\overset{\mathscr{E}_1'}{\nabla}}{\vdots}\,(e_1')}{\Omega,[\Psi,\Diamond B^\bullet,A^\circ]}\,(e_1) \qquad \dfrac{\overset{\mathscr{E}_2'}{\nabla}}{\Omega,[A^\bullet,\Psi,\Diamond B^\bullet,C^\bullet,D^\circ]}\,(e_2)}{\Omega,[\Psi,\Diamond B^\bullet,C\supset D^\circ]}\,(\mathsf{cut}^{[\cdot]}) \quad\rightsquigarrow\quad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\Psi,[B^\bullet],A^\circ}{\Psi,[B^\bullet],\Diamond B^\bullet,C^\bullet,A^\circ}\,(w^\bullet) \qquad \dfrac{A^\bullet,\Psi,\Diamond B^\bullet,C^\bullet,D^\circ}{A^\bullet,\Psi,[B^\bullet],\Diamond B^\bullet,C^\bullet,D^\circ}\,(w^+_{[\cdot]})}{\Psi,[B^\bullet],\Diamond B^\bullet,C^\bullet,D^\circ}\,(\mathsf{cut})}{\Psi,\Diamond B^\bullet,\Diamond B^\bullet,C^\bullet,D^\circ}\,(\Diamond_1^\bullet)}{\Psi,\Diamond B^\bullet,C^\bullet,D^\circ}\,(c^\bullet)}{\Omega,[\Psi,\Diamond B^\bullet,C\supset D^\circ]}\,(\supset_2^\circ)$$

(iv) A° is *side* in (e_1') and $A^\bullet$ is *principal* in (e_2). Similar to (iii), A cannot be a $\Box$-formula. For cases when $A=C\wedge D$ or $C\supset D$, the proof is the same to (iii), since the corresponding $\circ$-rules with respect to A° are invertible. Thus we only have two cases remaining:

- $A=\Diamond D$, in this case, the derivation $\mathscr{E}$ is unfolded as follows, where $\Psi=\Phi'$,

$$\dfrac{\dfrac{\overset{\mathscr{E}_1'}{\nabla}}{\vdots}\,(e_1')}{\Omega^*,[\Psi^*,\Diamond B^\bullet,\Diamond D^\circ]}\,(e_1) \qquad \dfrac{\overset{\mathscr{E}_2'}{\nabla}}{\Omega,[\Diamond D^\bullet,\Psi,\Diamond B^\bullet]}\,(e_2)}{\Omega,[\Psi,\Diamond B^\bullet]}\,(\mathsf{cut}^{[\cdot]})$$

We have the following cases:

(a) If $e_1'=\Diamond_2^\bullet$, the conclusion of cut follows from the premise of (e_1') by $(\Diamond_2^\bullet)$ twice.

(b) If e_1' is one of the $\bullet_1$-rules except $\Diamond_1^\bullet$, then from the premise of (e_1'), we apply (e_1) first, then $(\mathsf{cut}^{[\cdot]})$ to the conclusion of (e_1) with the conclusion of (e_2), and the $\bullet_2$-counterpart of (e_1') in the end.

(c) If e_1' is $\Diamond^\circ$, then $\Psi,[B^\bullet,D^\circ]$ is derivable, we transform the whole derivation into

$$\dfrac{\dfrac{\dfrac{\dfrac{\Psi,[B^\bullet,D^\circ]}{\Psi,\Diamond B^\bullet,[B^\bullet,D^\circ]}\,(w^\bullet) \qquad \dfrac{[D^\bullet],\Psi,\Diamond B^\bullet,F^\circ}{[D^\bullet,B^\bullet],\Psi,\Diamond B^\bullet,F^\circ}\,(w^\bullet)}{[B^\bullet],\Psi,\Diamond B^\bullet,F^\circ}\,(\mathsf{cut}^{[\cdot]})}{\Omega,[\Psi,\Diamond B^\bullet,\Diamond B^\bullet,F^\circ]}\,(\Diamond_2^\bullet)}{\Omega,[\Psi,\Diamond B^\bullet,F^\circ]}\,(c^\bullet)$$

(d) If e'_1 is one of the $\bullet_2$-rules or $\Diamond^\bullet_1$, let us consider the rules applied above e'_1. Assume that before any $\circ$-rule is applied, there are no $\bullet_1$-rules applied above e'_1, otherwise we return to case (a), switching the rule order and simulating the derivation with the $\bullet_2$-counterpart as above. Let us enumerate the branches of $\mathscr{E}'_1$ as $\mathscr{B}_1, \ldots, \mathscr{B}_k$ and for each $\mathscr{B}_i$ further enumerate sequents contained in it as $\Phi_{i_0}, \ldots, \Phi_{i_j}, \ldots, \Phi_{i_m}$ such that $\Phi_{i_0} = \Phi'^*, [B^\bullet], \Diamond D^\circ$ and Φ_{i_m} is an axiom. Then for each $\mathscr{B}_i$, it implies one of the following situations regarding the output formula:

(h1) $\Diamond D^\circ$ is in each Φ_{i_j}.

(h2) there is some $i_k \geq 1$ such that $\Phi_{i_k} = \Theta, [\Lambda], \Diamond D^\circ$ for some Θ, Λ and $\Phi_{i_{k+1}} = \Theta, [\Lambda, D^\circ]$; meanwhile, for any $i_j \leq i_k$, the output formula of Φ_{i_j} is $\Diamond D^\circ$.

(h3) there is some $i_k \geq 1$ such that $\Phi_{i_k} = \Theta, [\Psi, P \supset Q^\bullet], \Diamond D^\circ$ for some Θ, Ψ, P, Q and $\Phi_{i_{k+1}} = \Theta, [\Psi, P \supset Q^\bullet, P^\circ]$; meanwhile, for any $i_j \leq i_k$, the output formula of Φ_{i_j} is $\Diamond D^\circ$.

We process each $\mathscr{B}_i$ respectively and construct the corresponding $\mathscr{B}'_i$ for it. If (h1) holds, this means $\Diamond D^\circ$ is unused and hence side in the whole branch, so we can just replace each occurrence of D° in $\mathscr{B}_i$ with the output formula F° of Ψ^+ and then get $\mathscr{B}'_i$. If (h2) holds, we reconstruct $\mathscr{B}_i$ into $\mathscr{B}'_i$ as follows. First we keep all the components Φ_{i_j}'s for $j > k+1$. Next, we remove Φ_{i_k}; and as in (b), cut $\Phi_{i_{k+1}}$ with the premise of (e_2) and obtain Φ'_{i_k} as follows

$$\dfrac{\dfrac{\Theta, [\Lambda, D^\circ]}{\Psi, \Diamond B^\bullet, \Theta, [\Lambda, D^\circ]}\,(w^\bullet) \qquad \dfrac{[D^\bullet], \Psi^+, \Diamond B^\bullet}{[\Lambda, D^\bullet], \Psi^+, \Diamond B^\bullet, \Theta}\,(w^\bullet)}{\Psi^+, \Diamond B^\bullet, \Theta, [\Lambda]}\,(\mathsf{cut}^{[\cdot]})$$

Then for all the Φ_{i_j}'s with $j < k$, let $\Phi'_{i_j} = \Psi, \Diamond B^\bullet, \Phi^*_{i_j}, F^\circ$.

Lastly, if (h3) holds, to construct $\mathscr{B}'_i$ from the current $\mathscr{B}$, we keep all the components Φ_{i_j}'s for $j > k$ and for $j \leq k$, we replace $\Diamond D^\circ$ with F°, which is the output formula of the final conclusion of $(\mathsf{cut}^{[\cdot]})$. Thus, for instance, $\Phi'_{i_k} = \Theta, [\Psi, P \supset Q^\bullet], F^\circ$. Note that the way we process each branch $\mathscr{B}$ in fact eliminates the cut formula $\Diamond D^\circ$ in the whole branch by substituting with the output formula at certain points, but this does not influence other rule applications in the derivation, thus we can connect each $\mathscr{B}'_i$ and reconstruct the derivation with contraction ending with $\Psi^+, \Diamond B^\bullet, [B^\bullet]$. Then by $(\Diamond^\bullet_2)$ and contraction, we conclude $\Omega, [\Psi, \Diamond B^\bullet]$.

- $A = C \vee D$, this case goes similarly to the previous one when $A = \Diamond D$.

This completes the whole proof of cut-admissibility. □

Proposition 5.7. *If a tight sequent Ω is provable in $\mathsf{SC}^\times_{\mathbf{FIK}}$ then it has a* clean *proof.*

Proof. Assume Ω is derivable in $\mathsf{SC}_{\mathbf{FIK}}$ with a derivation $\mathscr{D}$, we prove by induction on the height h of $\mathscr{D}$ that $\mathscr{D}$ can be transformed into a derivation of Ω in $\mathsf{SC}^\times_{\mathbf{FIK}}$.

First $h = 0$, if a tight sequent is an initial sequent in $\mathsf{SC}_{\mathbf{FIK}}$, then itself is a clean proof. Next $h > 0$, let us consider rule application (r) that might produce non-tight sequents. We show the following cases and others are similar.

(1) $r = \wedge^\bullet_2$ or $r = \vee^\bullet_2$, we only show the case of conjunction.

$$\dfrac{\vdash^{n-1} \Omega', [\Phi, A \wedge B^\bullet, A^\bullet, B^\bullet], [\Phi, A \wedge B^\bullet, A^\bullet, B^\bullet]}{\vdash^{n} \Omega', [\Phi, A \wedge B^\bullet], [\Phi, A \wedge B^\bullet, A^\bullet, B^\bullet]}\,\wedge^\bullet_2$$

From the premise by contraction and weakening in $\mathsf{SC}^{\times}_{\mathbf{FIK}}$, we obtain the conclusion at a smaller height then we apply IH.
(2) $r = \Diamond^{\circ}$.

$$\frac{\vdash^{n-1} \Omega', [\Phi, A^{\circ}], [\Phi, \Theta]}{\vdash^{n} \Omega', [\Phi], [\Phi, \Theta], \Diamond A^{\circ}} \Diamond A^{\circ}$$

In this case, the premise has a block with output formula but its input part is included in some other block, so it is not tight. Assume w.l.o.g there are no other blocks such that Φ, Θ is a subset of it. By Proposition 5.4, we have $\vdash^{n-1} \Omega', [\Phi], [\Phi, \Theta, A^{\circ}]$, which is a tight sequent. By IH, we have a clean proof for it, then we add the step of $\Diamond A^{\circ}$ as

$$\frac{\vdash^{n-1} \Omega', [\Phi], [\Phi, \Theta, A^{\circ}]}{\vdash^{n} \Omega', [\Phi], [\Phi, \Theta], \Diamond A^{\circ}} \Diamond^{\circ}$$

and obtain a clean proof for the conclusion.
(3) $r = \Box^{\bullet}$.

$$\frac{\vdash^{n-1} \Omega', \Box A^{\bullet}, [\Phi, A^{\bullet}], [\Phi, A^{\bullet}, F^{\circ}]}{\vdash^{n} \Omega', \Box A^{\bullet}, [\Phi], [\Phi, A^{\bullet}, F^{\circ}]} \Box^{\bullet} \quad \text{and} \quad \frac{\vdash^{n-1} \Omega', \Box A^{\bullet}, [\Phi, A^{\bullet}], [\Phi, A^{\bullet}]}{\vdash^{n} \Omega', \Box A^{\bullet}, [\Phi], [\Phi, A^{\bullet}]} \Box^{\bullet}$$

In the first case, the premise is not tight for the same reason as (2). Both cases are similar to (1). By weakening and contraction we obtain the conclusion at a smaller height so that we can apply IH. $\Box$